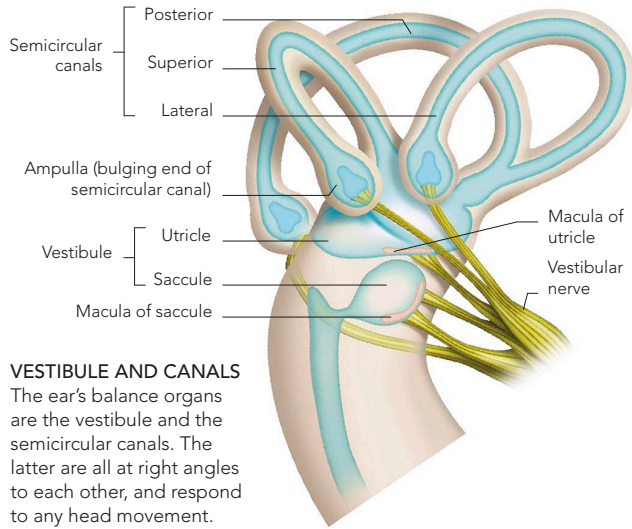


THE PROCESS OF BALANCE

Balance involves analysing sensory inputs from the eyes, skin, and muscles, and then adjusting the body's position through motor outputs. The vestibule and semicircular canals of the inner ear play a key role, too. The vestibule responds mainly to the position of the head relative to gravity (static equilibrium), while the canals react chiefly to the speed and direction of head movements (dynamic equilibrium).

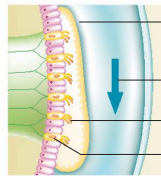
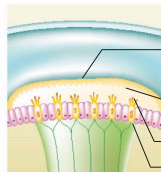


VESTIBULE AND CANALS

The ear's balance organs are the vestibule and the semicircular canals. The latter are all at right angles to each other, and respond to any head movement.

VESTIBULE

The vestibule's two parts, the utricle and saccule, each have a patch, the macula, containing hair cells. The tips of the cells extend into a membrane covered in heavy mineral crystals (otoliths). With the head level, the saccule's macula is vertical and the utricle's horizontal. As the head bends forward, the hair cells monitor the head's position in relation to the ground.

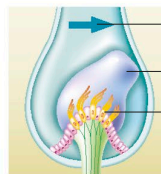
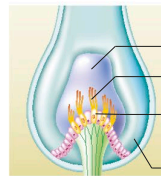


MACULA ACTION

Mineral crystals (otoliths) cover the membrane
Otolithic membrane
Hair of hair cell
Hair cell
Utricular macula rotated to vertical
Gravity pulls membrane
Hairs deflected
Hair cell stimulated

SEMICIRCULAR CANALS

Each semicircular canal has a bulge near one end, called the ampulla. This houses a low mound of hair cells, their hair ends set into a taller, jelly-like mound, the cupula. As the head moves, fluid in the canal lags behind, swirls past the cupula, and bends it. This pulls the hairs and triggers their cells to fire nerve signals.



AMPULLA ACTION

Cupula
Hairs of hair cells
Mound of hair cells (crista ampullaris)
Ampulla
Fluid swirls due to head motion
Cupula bends
Hair cells stimulated